

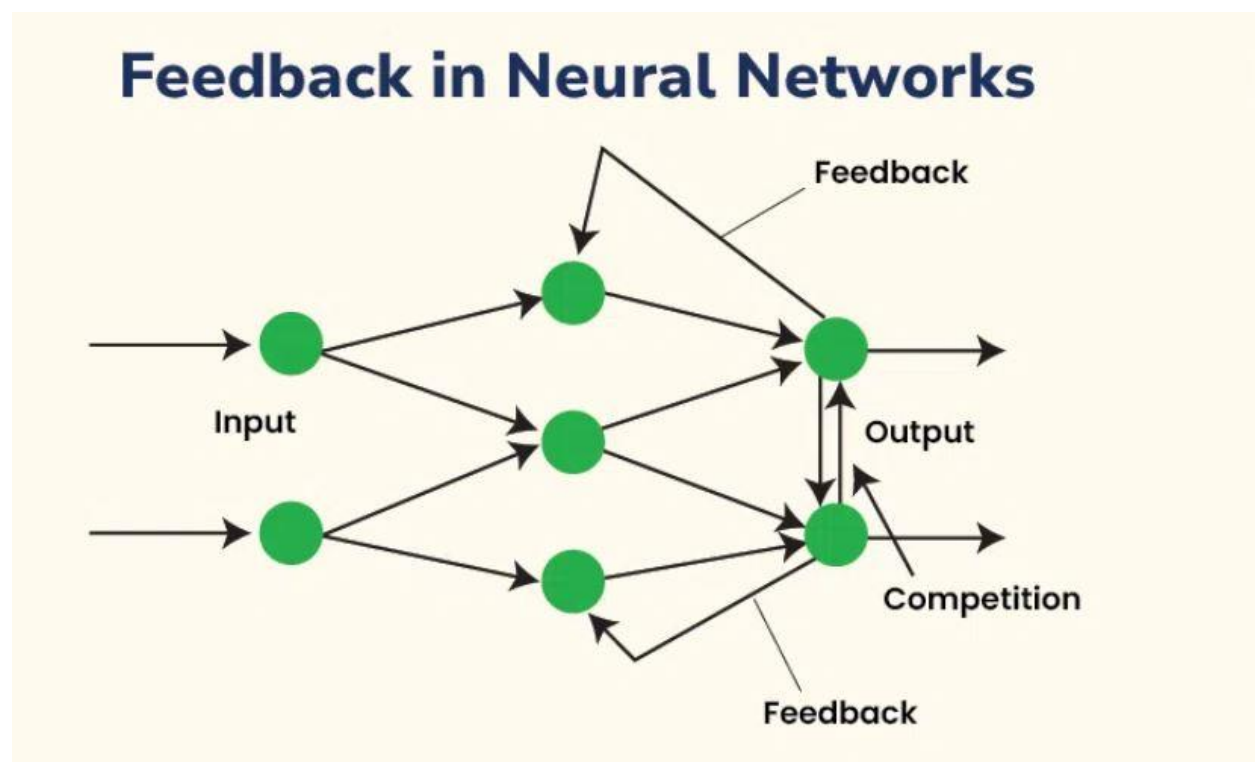
Feedback Neural Networks

These networks have connections that loop back, allowing information to be fed back into the network. This structure enables them to handle sequential data and temporal dependencies, making them suitable for tasks like time series prediction and language modeling.

Structure of Feedback Neural Networks

Feedback neural networks, or RNNs, are characterized by their ability to maintain a state that captures information about previous inputs. This is achieved through recurrent connections that loop back from the output to the input of the same layer or previous layers. The key components of an RNN include:

- **Input Layer:** Receives the input data.
- **Hidden Layers:** Contain neurons with recurrent connections that maintain a state over time.
- **Output Layer:** Produces the final output based on the processed information.



Mechanisms of Feedback in Neural Networks

- **Backpropagation:** Backpropagation is a method of feedback that involves the computation of the error gradient at each layer of the network. The error gradient is then used to update the network's parameters. Backpropagation is widely used in deep neural networks due to its efficiency and accuracy.
- **Recurrent Connections:** Recurrent connections involve the feedback of information from a later stage of the network to an earlier stage. This type of feedback is used in recurrent neural networks (RNNs), which are designed to handle sequential data.
- **Lateral Connections:** Lateral connections involve the feedback of information between neurons in the same layer. This type of feedback is used in applications such as image processing, where the goal is to capture spatial relationships between pixels.